

## GRUNDFOS MAGNA, GENI Module

Installation and operating instructions

GB D F I E P GR NL S FIN DK PL RU H  
SI HR YU RO BG CZ SK TR EE LT LV





# GRUNDFOS MAGNA, GENI Module

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Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

## 1. General description

The GENI module is an expansion module for circulator pumps GRUNDFOS MAGNA 25-60, 25-100, 32-60, 32-100, 40-100(D), 50-100.

The GENI module offers the following functions:

- **External start/stop**, see section 4.
- **External forced control**, see section 5.
- **External analog 0-10 V control**, see section 6.
- **Fault, ready and operating indication via signal relay**, see section 7.
- **Control of twin-head pumps**, see section 8.
- **Bus communication via GENIbus**, see section 9.

## 2. Connection

The electrical connection and protection must be carried out in accordance with local regulations.

The GENI module is powered wirelessly from the pump. To ensure correct function, follow the instructions in section 3. *Mounting the module.*

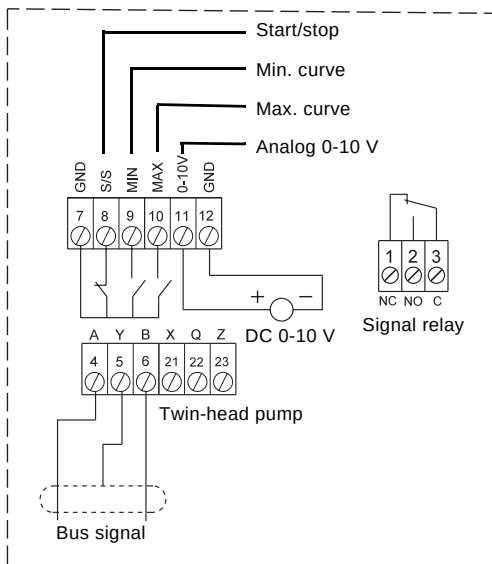
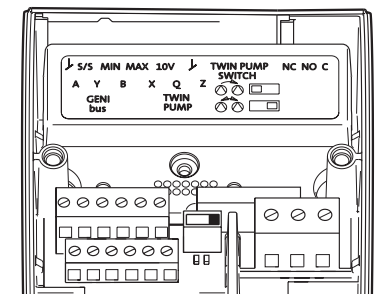
**Note:**

- All cables must be heat-resistant up to +85°C.
- All cables must be installed in accordance with EN 60204-1.



Wires connected to

- outputs NC, NO, C and
  - inputs Start/stop, A, Y, B, MIN, MAX, 10 V, X, Q, Z
- must be electrically isolated from each other.



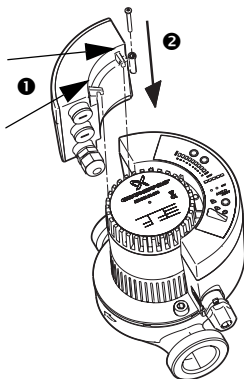
**Fig. 1** Connection terminals

See connection examples on pages 376 to 379.

### 3. Mounting the module

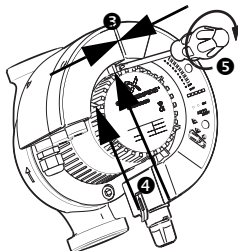
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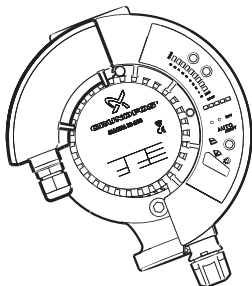
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1. Localise the two hooks on the module.
2. Gently push the module into place so that the hooks engage with the control box.



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3. Check that the module and the control box bear against each other.
4. Check that the two hooks of the module engage correctly with the control box.
5. Fasten the module with the screw.



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6. Correctly mounted module.

## 4. External start/stop

The GENI module incorporates a digital input for an external contact. The pump can be started and stopped via this input.

When started, the pump will operate with the setpoint set on the control panel or with the R100.

**Note:** If the external contact is used, the connection between terminals 7 and 8 must be replaced by the external connection.

### Functional diagram: Start/stop input

| Contact position                                                                  | Function      |
|-----------------------------------------------------------------------------------|---------------|
|  | Stop of pump  |
|  | Start of pump |

## 5. External forced control

The GENI module incorporates inputs for external signals for the forced-control functions:

- Max. curve duty.
- Min. curve duty.

See connection examples on pages 376 to 379.

### Functional diagram: Max. curve input

The max. curve input is only active if terminals 7 and 8 of the start/stop input are connected.

| Max. curve                                                                          |                                                                                     |             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------|
|  |  | Normal duty |
|  |  | Max. curve  |

### Functional diagram: Min. curve input

The min. curve input is only active if

- terminals 7 and 8 of the start/stop input are connected and
- terminals 7 and 10 of the max. curve input are not connected.

| Min. curve                                                                        |                                                                                   |             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------|
|  |  | Normal duty |
|  |  | Min. curve  |

## 6. External analog 0-10 V control

The GENI module has an input for an external 0-10 VDC analog signal transmitter. Via this input, the pump can be controlled by an external controller if the pump has been set to one of the following control modes:

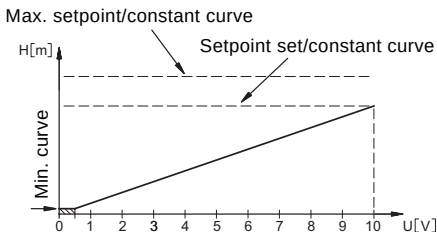
### • Constant curve

The external analog signal will control the pump curve within the range from the min. curve to the constant curve selected according to the characteristic in fig. 2.

### • Proportional- or constant-pressure control

The external analog signal will control the setpoint for the pump head between the setpoint corresponding to the min. curve and the setpoint selected according to the characteristic in fig. 2.

At an input voltage lower than 0.5 V, the pump will operate according to the min. curve. The setpoint cannot be changed.  
The setpoint can only be changed when the input voltage is higher than 0.5 V.



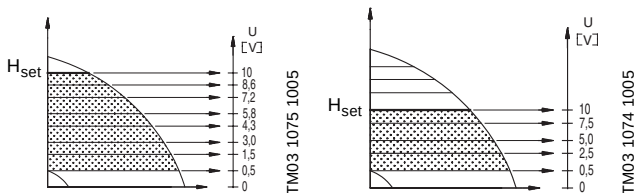
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**Fig. 2** Example of 0-10 V control

**Note:**

- Terminals 7 and 10 of the max. curve input must not be connected.
- Terminals 7 and 9 of the min. curve input must be connected.
- Terminals 7 and 8 of the start/stop input must be connected.

See connection examples on pages 376 to 379.



**Fig. 3** Example of analog control voltage with the pump in constant-pressure control mode

**Note:** As will appear from fig. 3, the control range will be limited if the set-point set,  $H_{set}$ , is reduced.

## 7. Signal relay

The function of the signal relay can be set with the R100.

Possible functions:

- *Fault*
- *Ready*
- *Operation.*

### 7.1 Fault indication

The signal relay is activated in case of

- *Pump blocked*
- *Internal fault*
- *Undervoltage.*

See section 7.4 *Functional diagram.*

When the fault disappears, the signal relay is deactivated. The fault cause is stored in the pump alarm log. The latest five faults can be called up with the R100.

**Note:** In case of a fault, the red indicator light on the control panel is on.

See installation and operating instructions for GRUNDFOS MAGNA 25-60, 25-100, 32-60, 32-100, 40-100(D), 50-100.

## 7.2 Ready indication

The signal relay is active when the pump is in operation or ready for operation.

See section 7.4 *Functional diagram*.

## 7.3 Operating indication

The signal relay is active as long as the pump is operating. If the pump is stopped on the control panel, with the R100 or because of a fault, the signal relay is deactivated and consequently gives a signal to an external control system, e.g. a building management system.

See section 7.4 *Functional diagram*.

## 7.4 Functional diagram

| Setting          | Relay                    | Description                         |
|------------------|--------------------------|-------------------------------------|
| <b>Fault</b>     | <p>1 2 3<br/>NC NO C</p> | Pump faulty                         |
| <b>Ready</b>     | <p>1 2 3<br/>NC NO C</p> | In operation<br>Ready for operation |
| <b>Operation</b> | <p>1 2 3<br/>NC NO C</p> | In operation                        |
| <b>-</b>         | <p>1 2 3<br/>NC NO C</p> | Relay dead                          |

## 8. Control of twin-head pumps

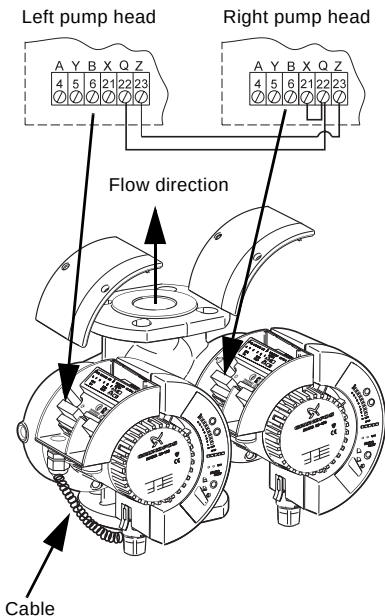
When fitted with two GENI modules, a twin-head pump functions fully automatically.

Mount a GENI module on the control box of each pump head and connect the modules using a cable.

Both pump heads must be connected to the electricity supply.

**Note:** Connect a short wire between the terminals 21 "X" and 22 "Q" in the module on the right pump head, see pages 378 and 379.

For the connection of the left and right pump heads, see pages 378 and 379.



**Fig. 4** Twin-head pump

It is possible to connect two single-head pumps so that they operate as a twin-head pump. In that case, it may be necessary to install non-return valves. It is advisable to contact Grundfos to ensure optimum operation.

The pump has these operating modes:

- **Alternating operation**

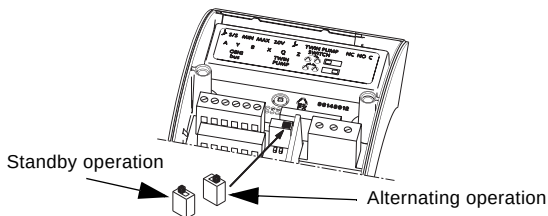
Pump operation alternates every 24 hours. If the duty pump stops due to a fault, the other pump will start.

- **Standby operation**

One pump is operating continuously. In order to prevent seizing-up, the other pump will start at a fixed frequency. If the duty pump stops due to a fault, the other pump will start.

Select the operating mode by means of the mechanical contact in each module.

| Operating mode | Left pump head | Right pump head |
|----------------|----------------|-----------------|
| Alternating    | Alternating    | Alternating     |
| Standby        | Alternating    | Standby         |
| Standby        | Standby        | Alternating     |
| Standby        | Standby        | Standby         |



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**Fig. 5** Mechanical contact

### Operating the pump

The connected pumps can be set and operated in the same way as single-head pumps. The duty pump uses its setpoint setting, whether it was made on the control panel, with the R100 or via bus.

**Note:** Both pumps should be set to the same setpoint and control mode. Different settings will result in different operation when changing between the two pumps.

### Checking the function

One pump must be operating and the other must be stopped.

- Start the pump that was stopped.
- Check that the operating pump stops.

**Note:** The check can be made, whether standby or alternating operation has been selected.

## 9. Bus communication via GENibus

The GENI module enables serial communication via an RS-485 input. The communication is carried out according to the Grundfos bus protocol, GENI-bus, and enables connection to the GRUNDFOS Pump Management System 2000, a building management system or another type of external control system.

Via the bus signal, it is possible to set pump operating parameters, such as desired setpoint, temperature influence, operating mode, etc.

At the same time, the pump can provide status information about important parameters, such as actual head, actual flow, power input, fault indications, etc.

For further details, consult the operating instructions for the GRUNDFOS Pump Management System 2000 or contact Grundfos.

**Note:** When the pump is controlled via a bus signal, the number of settings available on the pump control panel or with the R100 will be reduced.

## 10. Priority of settings

The external forced-control signals will influence the settings available on the pump control panel and with the R100. However, the pump can always be set to max. curve duty or to stop on the control panel or with the R100.

If two or more functions are activated at the same time, the pump will operate according to the setting with the highest priority.

The priority of the settings is as shown in the table:

| Priority | Possible settings          |                  |                  |
|----------|----------------------------|------------------|------------------|
|          | Pump control panel or R100 | External signals | Bus signal       |
| 1        | Stop                       | –                | –                |
| 2        | Max. curve                 | –                | –                |
| 3        | –                          | Stop             | Stop             |
| 4        | –                          | Max. curve       | Max. curve       |
| 5        | Min. curve                 | Min. curve       | Min. curve       |
| 6        | Setpoint setting           | –                | Setpoint setting |

|  |
|--|
|  |
|  |

Not active when the pump is controlled via bus.

Only active when the pump is controlled via bus.

**Example:** If the pump has been forced to stop via an external signal, the pump control panel, the R100 or a bus signal can only set the pump to max. curve.

As illustrated in the table, the pump does not react to external signals (max. curve and min. curve) when the pump is controlled via bus. If the pump is to react to external signals (max. curve and min. curve), the system must be configured for that function.

## 11. Fault finding chart

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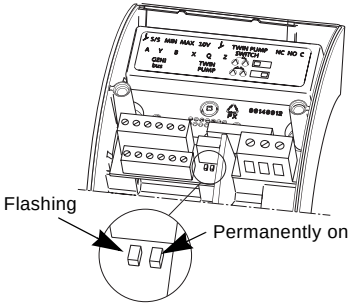


Before starting any work on the module, make sure that the electricity supply to the pump and the module has been switched off and that it cannot be accidentally switched on.

| Fault                                                              | Cause                                                                                  | Remedy                                                                                   |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| The pump does not react to the input start/stop, MIN, MAX or 10 V. | The wires are not connected correctly to the terminal block.                           | Connect the wires correctly.                                                             |
|                                                                    | The pump was stopped on the control panel or with the R100.                            | Set the pump back to normal duty.                                                        |
|                                                                    | GENI module not mounted correctly.<br>GENI module defective.<br>Control box defective. | Mount the GENI module correctly.<br>Replace the GENI module.<br>Replace the control box. |
| The pump does not send the correct signal from the signal relay.   | The wires are not connected correctly to the terminal block.                           | Connect the wires correctly.                                                             |
|                                                                    | Signal relay not configured correctly.                                                 | Configure the signal relay correctly.<br>The R100 can be used.                           |
|                                                                    | GENI module not mounted correctly.<br>GENI module defective.<br>Control box defective. | Mount the GENI module correctly.<br>Replace the GENI module.<br>Replace the control box. |

| Fault                                          | Cause                                                                                  | Remedy                                                                                   |
|------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| The pump does not react to the GENIbus signal. | The wires are not connected correctly to the terminal block.                           | Connect the wires correctly.                                                             |
|                                                | The pump was stopped on the control panel or with the R100.                            | Set the pump back to normal duty.                                                        |
|                                                | GENI module not mounted correctly.<br>GENI module defective.<br>Control box defective. | Mount the GENI module correctly.<br>Replace the GENI module.<br>Replace the control box. |

## 11.1 Checking the module

| Illustration                                                                       | Action                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ol style="list-style-type: none"> <li>1. Remove the module cover.</li> <li>2. Check the LEDs. When the pump is switched on and the module is correctly mounted, the left LED must be flashing and the right LED must be permanently on.</li> <li>3. Refit the module cover.</li> </ol> |

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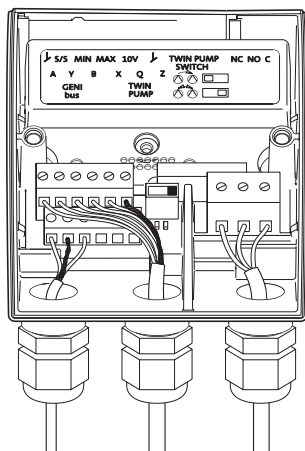
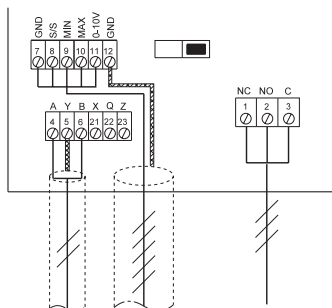
## 12. Technical data

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|                                                |                                                                                                                                                        |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs for max. and min. curves and start/stop | External potential-free contact.<br>Contact load: 5 V, 1 mA.<br>Screened cable.<br>Loop resistance: Maximum 130 $\Omega$ .                             |
| Input for analog 0-10 V signal                 | External signal: 0-10 VDC.<br>Maximum load: 1 mA.<br>Screened cable.                                                                                   |
| Input for control of twin-head pumps           | Screened cable.<br>Wire cross section: 0,25 - 1 mm <sup>2</sup> .<br>Cable length: Maximum 1 m.                                                        |
| Bus input                                      | Grundfos bus protocol, GENibus protocol, RS-485.<br>Screened cable.<br>Wire cross section: 0.25 - 1 mm <sup>2</sup> .<br>Cable length: Maximum 1200 m. |
| Output for signal relay                        | Potential-free changeover contact.<br>400 VAC, 6 A AC1.<br>30 VDC, 6 A.                                                                                |

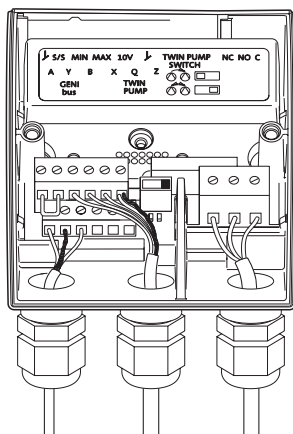
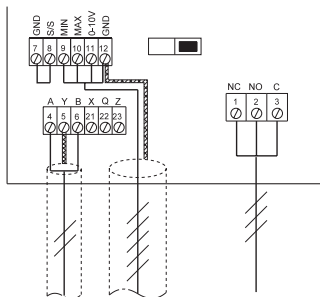
Subject to alterations.

## Connection example



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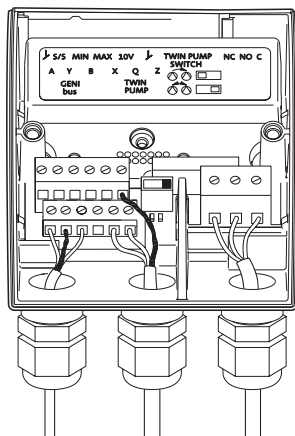
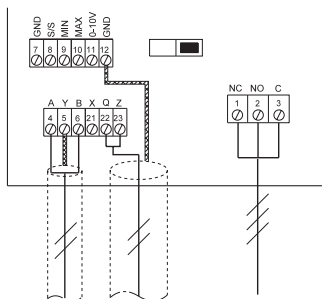
## Connection example



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# MAGNA D 40-100 twin-head pump

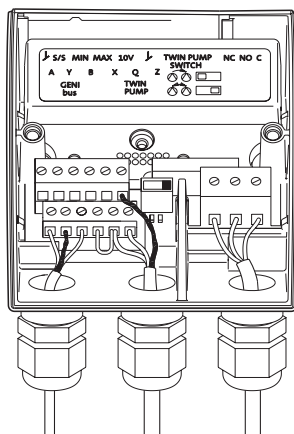
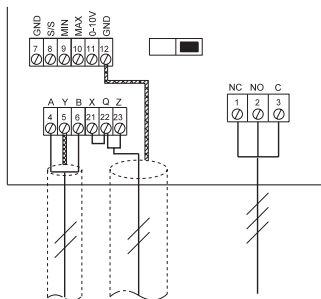
Left pump head when seen in the liquid flow direction. See fig. 4.



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## MAGNA D 40-100 twin-head pump

Right pump head when seen in the liquid flow direction. See fig. 4.



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Being responsible is our foundation  
Thinking ahead makes it possible  
Innovation is the essence

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